Lead 16.5: class-wide multi-target rho, measured

Round 4 of the isogeny-fibers study · 2026-10-11 · native Rust: `examples/r4_16_5_class_multitarget_rho.rs`
Records: `run_r28.json`, `run_r32.json`, `run_r36.json`, `fit.json` · full note: `RESULTS.md`

Labels. Values in the arm columns are **measured** operation counts. The reference columns (KS, $L\sqrt{(\pi/2)}$) are **derived**. Toy classes only (r from $2^{28.8}$ to $2^{37.2}$). No sub-rho claim. No claim about a deployed curve. Wall time is not a metric here.

1. Question and falsifier

§14.5 predicts the following. Put L keys on L curves of one isogeny class, linked by explicit small-degree isogenies. Transport them to one curve. Solving all L then costs what L-target (Kuhn–Struik) rho costs on a single curve: about $\sqrt{(2Lr)}$ in total, against $L\cdot\sqrt{(\pi r/2)}$ independently.

Falsifier, fixed in advance. Arm B transports the targets and runs a shared-table multi-target rho on E_0. Arm C plants the L targets on E_0 directly and runs the same rho. The transport argument fails if B is not within sampling error of C.

2. What ran

- **Class.** E_0: $y^2 = x^3 + ax + b$ over F_p, with $\#E_0 = h\cdot r$, r prime and $6 \mid h$.
  - A native BFS over rational 2-isogenies and 3-isogenies (Vélu) found 65 curves, each with a distinct j-invariant.
  - Every edge has a certified dual: $\text{dual}\circ\varphi = [\ell]$ on the generator.
  - Each 3-isogeny codomain was cross-checked against `crypto_lib`'s `velu_codomain`.
  - Generators: $P_i = \psi_i(P_0)$. The dual path then gives $[d_i]P_0$.
- **Arms.** Targets sit on $E_{1..L}$.
  - **A:** L independent DP-rho runs, one per curve.
  - **B:** transport to E_0, then one multi-target DP rho that keeps its table across targets. Each log is mapped back by d_i^{-1} and verified on E_i.
  - **C:** L targets on E_0, same rho.
  - All arms use the same rho: a target-independent 32-adding walk with DP probability $2^{-\text{dpbits}}$.
  - $L \in \{1, 4, 16, 64\}$. There are 100, 40 and 20 trials at $r = 2^{28.79}$, $2^{31.40}$ and $2^{37.18}$.
  - All 40,800 recovered logs were verified.
- **Unit.** S = total group operations / $\sqrt{r}$. The total includes the step-table and start scalar multiples. Transport is counted in field M and I and added to B.

3. Result

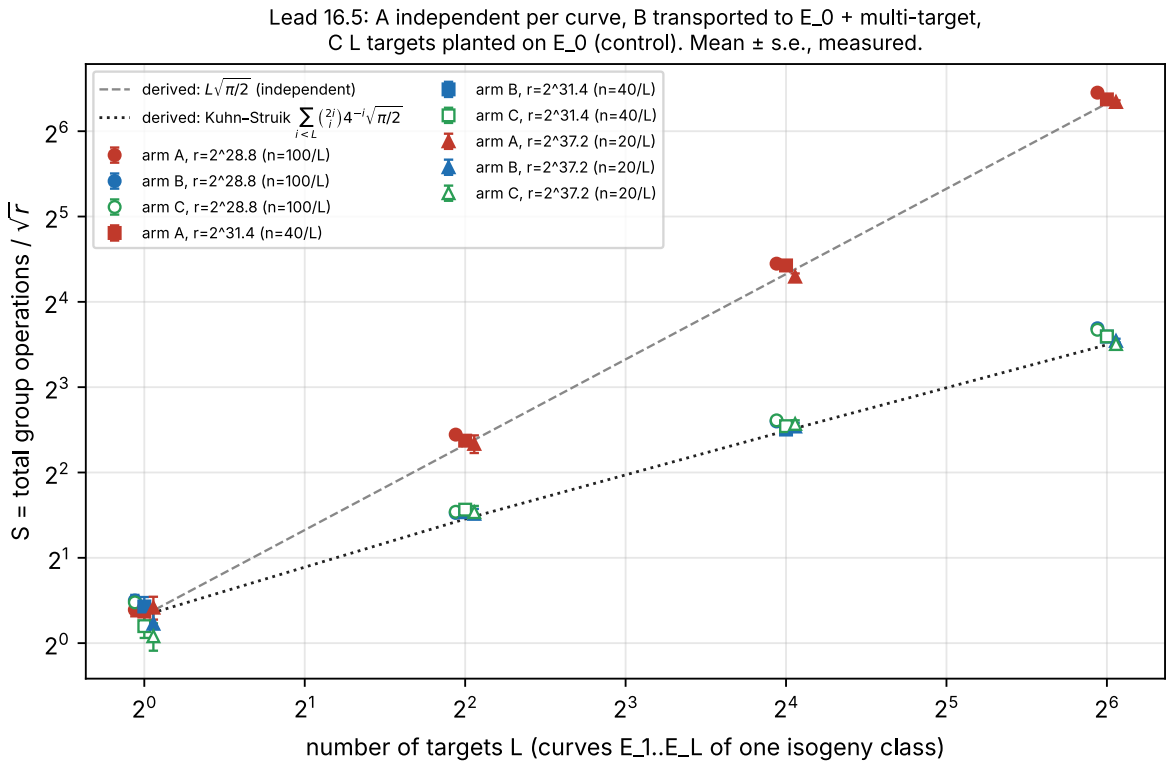


Figure 1: Measured S (mean ± s.e.) against L for the three arms at three sizes of r. The dashed line is derived independent rho, $L\sqrt{(\pi/2)}$. The dotted line is the derived Kuhn–Struik reference. B and C lie on top of each other along the √L-type line. A follows the linear line. Data: `run_r28.json`, `run_r32.json`, `run_r36.json`.

r	L	A	B	C	B/C [95%]	transport/target
$2^{28.79}$	1	1.312	1.414	1.393	1.015 [0.879, 1.152]	9.0 M + 1.0 I
$2^{28.79}$	64	87.543	12.909	12.752	1.012 [0.997, 1.028]	78.8 M + 8.8 I
$2^{31.40}$	1	1.292	1.347	1.148	1.173 [0.894, 1.452]	9.0 M + 1.0 I
$2^{31.40}$	64	82.869	12.062	12.084	0.998 [0.972, 1.025]	34.6 M + 3.8 I
$2^{37.18}$	1	1.334	1.170	1.057	1.107 [0.753, 1.461]	9.0 M + 1.0 I
$2^{37.18}$	64	81.143	11.648	11.354	1.026 [0.979, 1.073]	78.8 M + 8.8 I

Derived references: KS is 1.253 at L = 1 and 11.292 at L = 64. $L\sqrt{(\pi/2)}$ is 80.21 at L = 64. The L = 4 and L = 16 rows are in `RESULTS.md`; there too every B/C interval contains 1.

Fits (`fit.json`). These are least squares of ln S on ln L, with 95% intervals from a trial bootstrap.

- β_A is 1.009, 1.003 and 0.987 at the three sizes of r.
- β_B is 0.532 [0.513, 0.554], 0.523 [0.489, 0.559] and 0.548 [0.504, 0.604].
- β_C is 0.533, 0.558 and 0.565.
- The KS reference over the same L gives a derived 0.53.

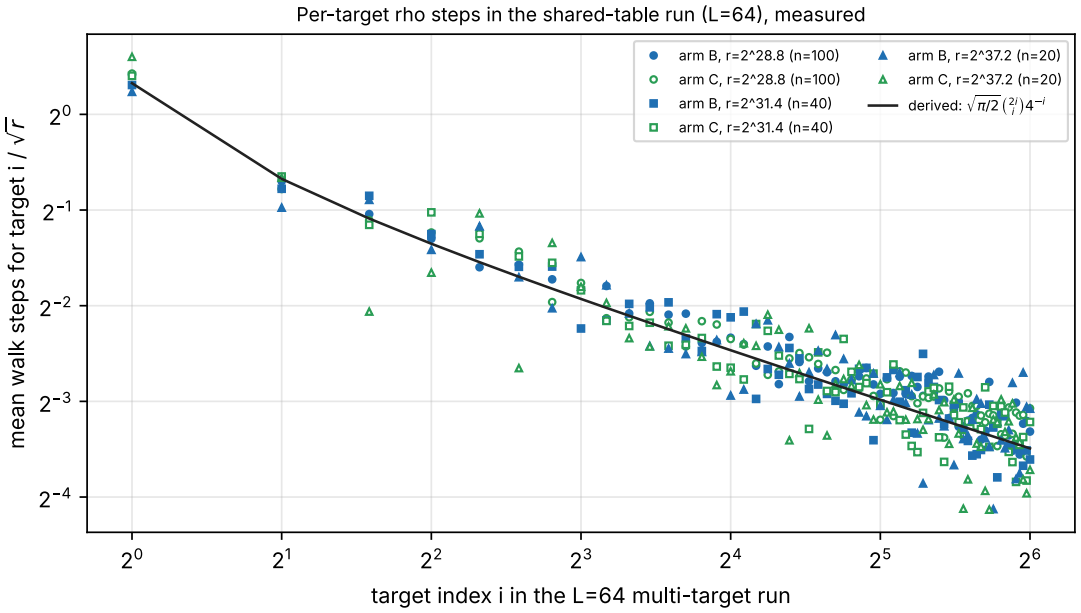


Figure 2: Mean rho steps / √r spent on the i-th target in the L = 64 shared-table runs (measured). B is transported; C is native. The solid line is the derived Kuhn–Struik per-target expectation.

4. Verdict: confirmed on toy classes

- **The falsifier did not trigger.** B/C contains 1 in all 12 (r, L) cells. At L = 64 the ratios are 1.012, 0.998 and 1.026.
- **Saving.** The class-wide batch is 6.8–7.0× cheaper than per-curve solving at L = 64. The derived ratio is 7.1.
- **Transport is additive and negligible.** It is 9 M + 1 I per isogeny step, with depth at most 17.
- **Gap to the ideal.** B and C sit 3–14% above the ideal KS curve, by the same amount in both arms. The cause is accounted setup and DP tails. This is engineering, not a separation.
- **Protocol reading.** Per-user curves of one order, linked by small rational isogenies, give no multi-target protection beyond a single shared curve. This is the multi-target face of §13.6.

5. Limits

- One class per size of r, with L ≤ 64.
- Only rational 2-isogeny and 3-isogeny edges; path finding is not priced.
- The rho is the example's own DP rho. ecbench does not pool multi-target runs, so there is no replay certificate. It has no negation map, which would scale all arms alike.
- The r = $2^{28.79}$ run was replayed with the final binary and came out bit-identical.

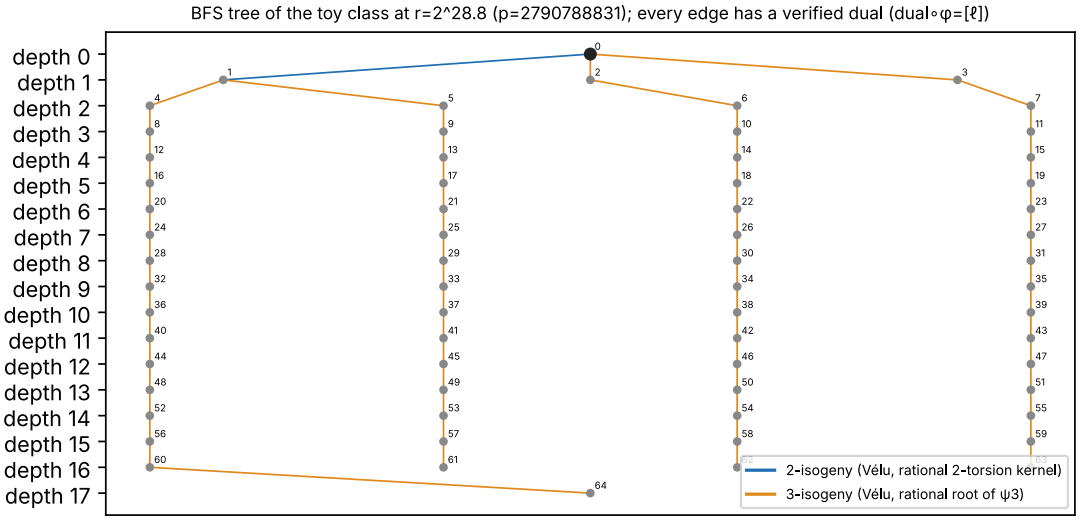


Figure 3: BFS tree of the r = $2^{28.79}$ class, drawn from the recorded parents and edge degrees. Blue edges are 2-isogenies; orange edges are 3-isogenies. Node labels are curve indices in `run_r28.json` `class.curves`.